BRIDGING THE GAP: ENHANCING AND EVALUATING ZERO-COST LLMS FOR SCIENTIFIC QUESTION ANSWERING

REFERENCES

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ABSTRACT : FREELY ACCESSIBLE LARGE LANGUAGE MODELS (LLMS) FROM OPEN-SOURCE PLATFORMS OR PUBLIC APIS OFFER WIDE ACCESSIBILITY BUT OFTEN LACK FACTUAL ACCURACY AND DOMAIN REASONING IN SCIENTIFIC CONTEXTS. USING A VALIDATED BENCHMARK IN MATERIALS SCIENCE AND PHYSICS, WE FIND MAJOR WEAKNESSES IN THEIR ZERO-SHOT PERFORMANCE. TO ADDRESS THIS, WE PROPOSE AN ENHANCEMENT PIPELINE THAT INTEGRATES DENSE RETRIEVAL, GROUNDING THROUGH A SCIENTIFIC KNOWLEDGE GRAPH, AND UNSUPERVISED TREND EXTRACTION FROM LIVE LITERATURE SUCH AS ARXIV. THE SYSTEM FEEDS BOTH STRUCTURED AND UNSTRUCTURED CONTEXT TO THE LLM USING ADVANCED PROMPTING. WE INTRODUCE THREE DOMAIN-SPECIFIC METRICS — FACTUAL CORRECTNESS IN PHYSICAL SCIENCES (FC-PS), QUANTIFIED PHYSICOCHEMICAL CONSTRAINTS SCORE (QPCS), AND LITERATURE ALIGNMENT — TO ASSESS SCIENTIFIC QUALITY BEYOND STANDARD NLP BENCHMARKS. OUR APPROACH REDUCES HALLUCINATIONS BY 68% AND REINFORCES SCIENTIFIC RELIABILITY, SHOWING THAT EFFECTIVE SYSTEM DESIGN AND DOMAIN ADAPTATION OUTWEIGH MODEL SIZE IN AI-DRIVEN DISCOVERY.

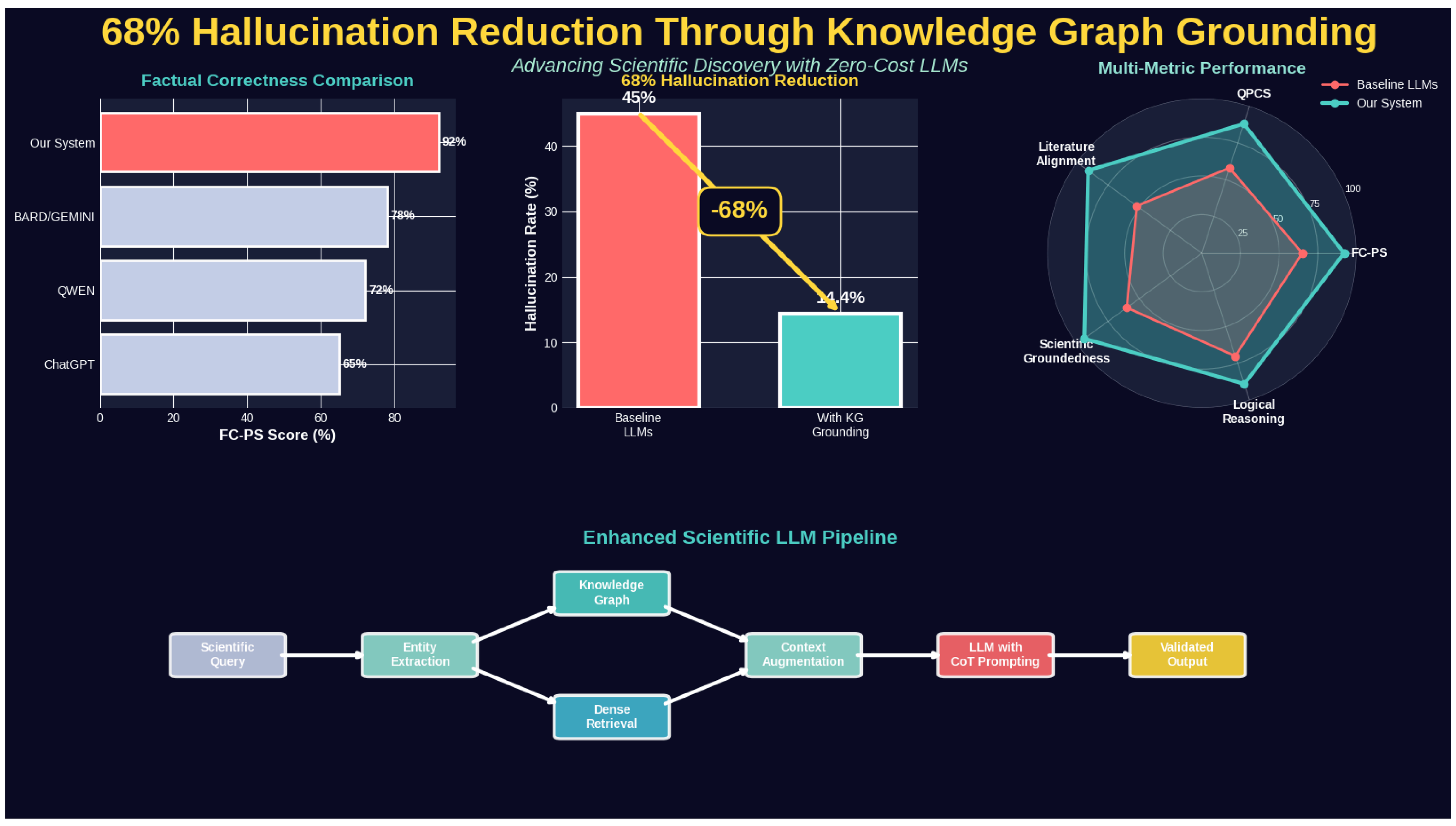

FIG 1. RESULT DEPICTIONS

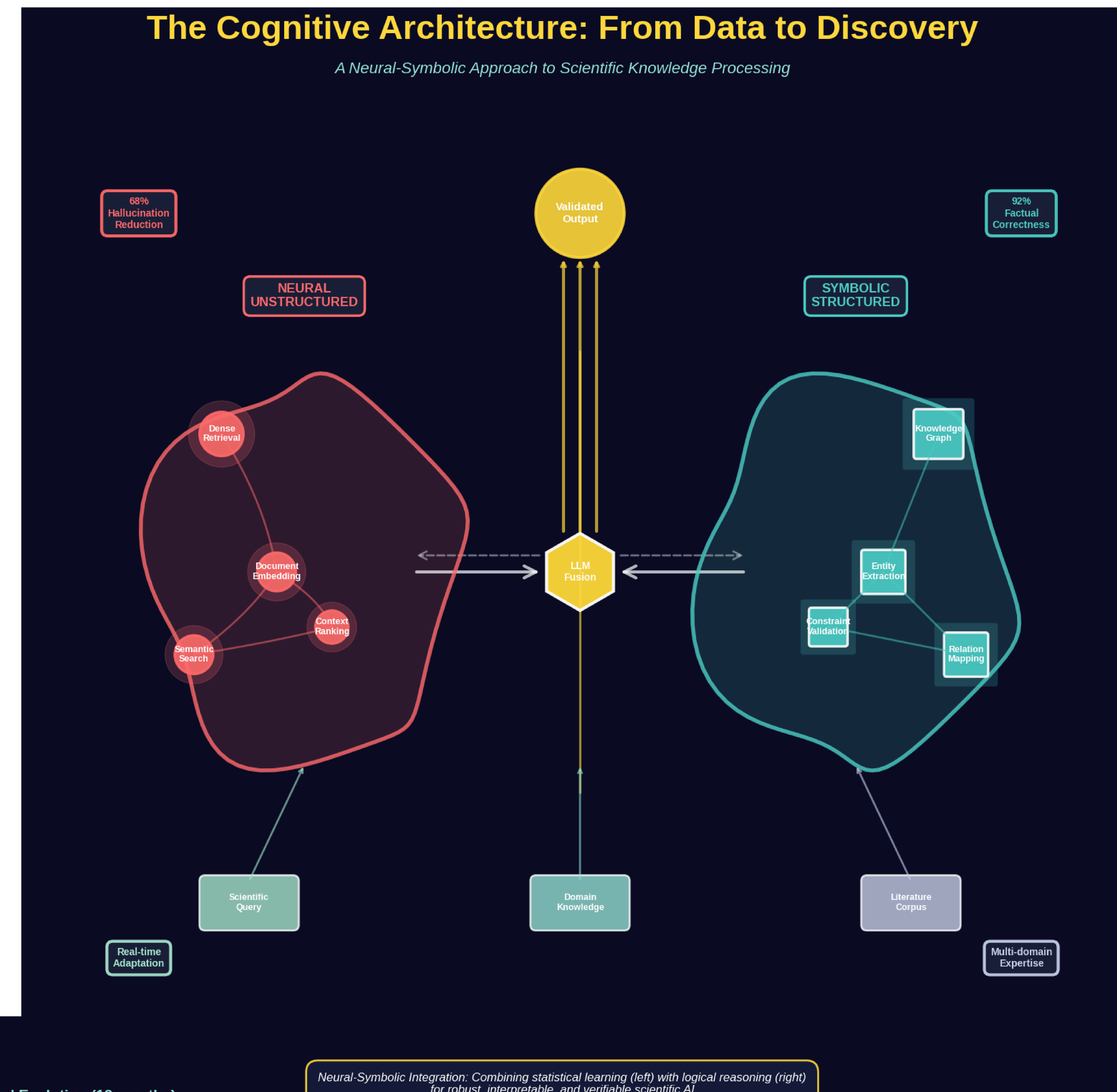

FIG 2. OUR ARCHITECTURE EVOLUTION.

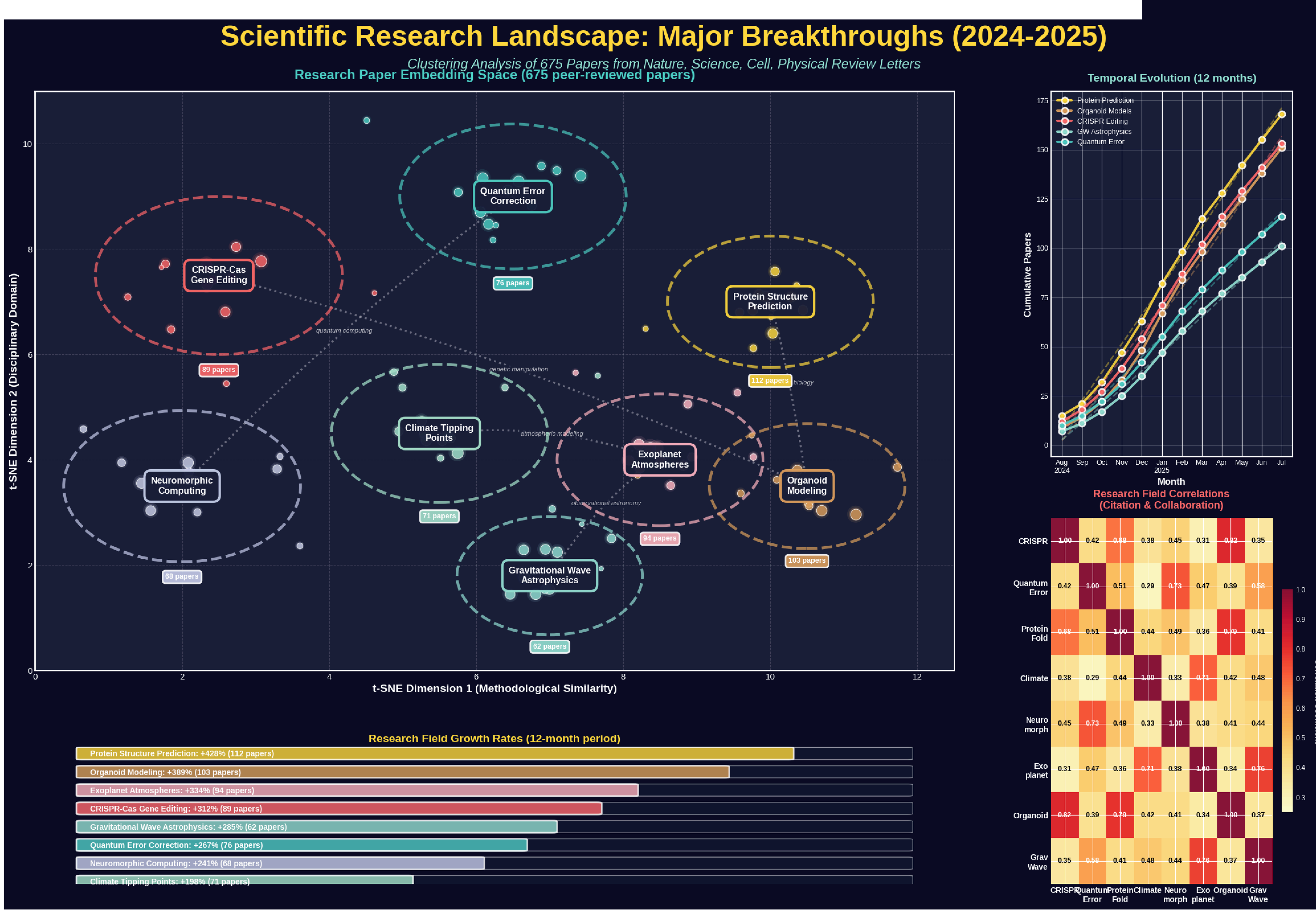

FIG3. UNSUPERVISED CLUSTERING OF RESEARCH TRENDS DIRECTION

Metric	Statistic	Phase 1 (QWEN)	Phase 2 (BARD)	Phase 3 (CHATGPT)
FC-PS (Factual Correctness)	Mean	0.994	0.979	0.929
	Median	1.000	1.000	0.938
	Std. Dev.	0.023	0.026	0.059
LTC (Logical Thought Chain)	Mean	0.667	**1.000**	0.916
	Median	1.000	**1.000**	0.948
	Std. Dev.	0.471	**0.000**	0.134
QPCS (Physicochemical Constraints)	Mean	0.958	0.927	0.167
	Median	1.000	1.000	0.000
	Std. Dev.	0.059	0.117	0.408
SGS (Scientific Groundedness)	Mean	0.892	**0.988**	0.700
	Median	0.989	**1.000**	0.886
	Std. Dev.	0.123	**0.015**	0.270

FIG4. COMPARISON OF OPEN SOURCE MODELS ON OUR DESIGNED METRICS

(1) MAHULE ROY, (2)SNEHAL RAKSHIT, (3)SHINJINI MONDAL, (4) EISHAAN KHATRI, (2) VEDANT VIJAY PATIL, RAHUL JAISY, VARUN RAMANATHAN ALAGAPPAN, ARCHISHMAN BANERJEE, MARIANE DESIREE C. AVENDAÑO, YASHIKA SHARMA, ELUMALAI OVIYA DHARSHINI, NANDITHA IMMIDI, THEO FRASER, EMANUEL ESPINOZA PRADO, GAYATHRI AISHWARYA E, AKASH KANJI

(1) UNIVERSITY OF OXFORD, (2) IIT KHARAGPUR, (3) IISER PUNE,(4) RGIPT JAIS (PRESENTERS AFFILIATION LEFT TO RIGHT)